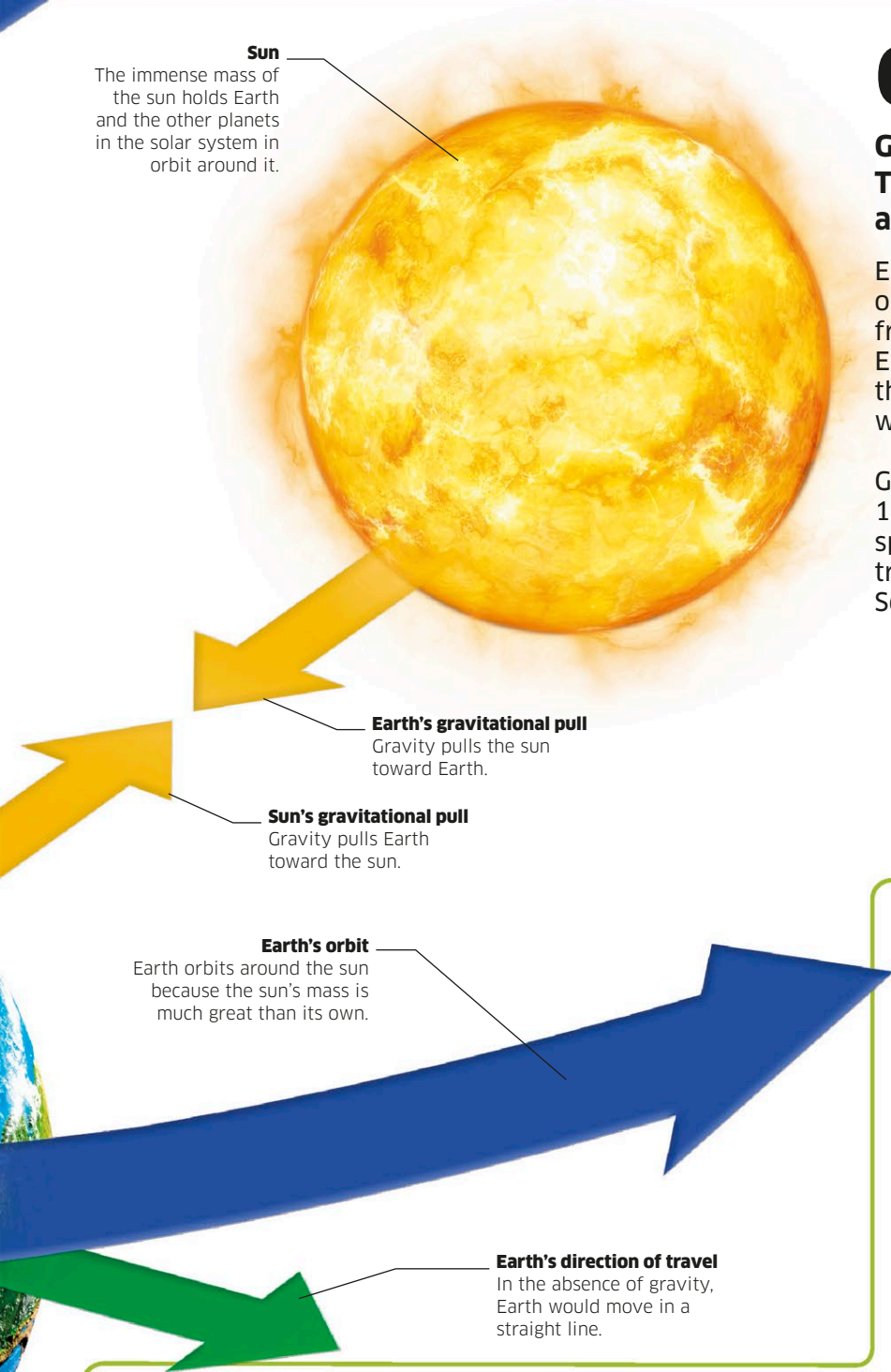


Gravity

Gravity is a force of attraction between two objects. The more mass the objects have and the closer they are to each other, the greater the force of attraction.

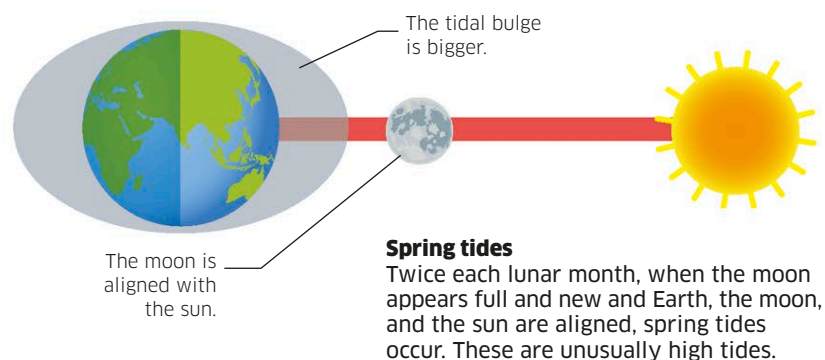
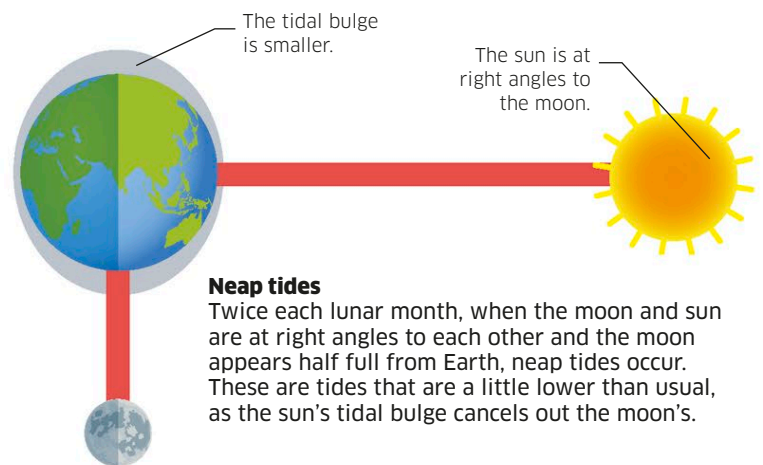
Earth's gravity is the gravitational force felt most strongly on the planet: it is what keeps us on the ground and keeps us from floating off into space. In fact, we pull on Earth as much as Earth pulls on us. Gravity also keeps the planets in orbit around the sun, and the moon around Earth. Without it, each planet would travel in a straight line off into space.

The best way scientists can explain gravity is with the General Theory of Relativity, formulated by Albert Einstein in 1915. According to this theory, gravity is actually caused by space being distorted around objects with mass. As objects travel through the distorted space, they change direction. So, according to Einstein, gravity is not a force at all!



Tides

The gravitational pull of the moon and the sun cause the oceans to bulge outward. The moon's pull on the oceans is strongest because it is closest to Earth, and it is the main cause of the tides. However, at certain times of each lunar month, the sun's gravity also plays a role, increasing or decreasing the height of the tides.



Mass and weight

Mass is the amount of matter an object contains, which stays the same wherever it is. It is measured in kilograms (kg). Weight is a force caused by gravity. The more mass an object has and the stronger the gravity, the greater its weight. Weight is measured in newtons (N).

